Part II : Eigenface Feature Detection

Mathematical Explanation

Eigenface generation

1. Present the mathematical explanation of eigenfaces, including how to generate the eigenfaces from the data set.

Moustache Detection Functionality

2. Present the mathematical explanation of how to build a rudimentary moustache detector by projecting onto -dimensional "eigenface space".

Moustache Detection Efficacy

Figure 1: 30 moustache detected faces

The system achieved an approximate accuracy of 89.3%, correctly classifying the majority of images with clear moustache features. However through initial inspection the majority of errors come from false positives from the variance in lighting conditions between different images.

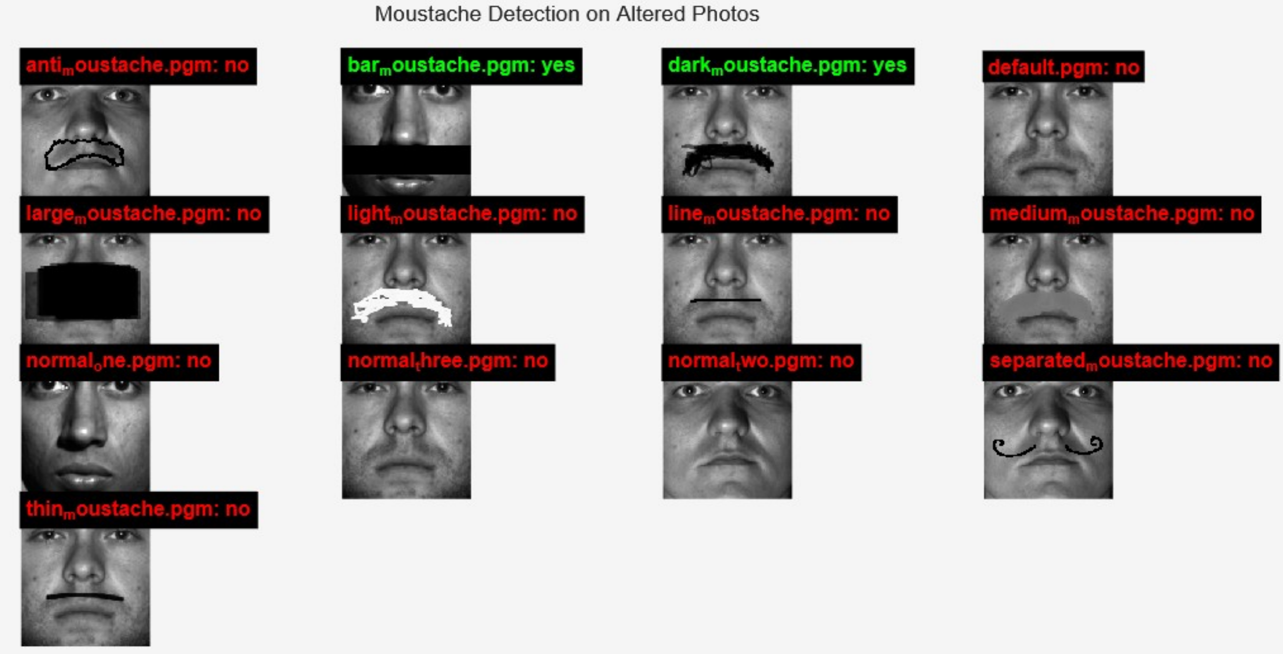

Figure 2: 13 altered faces tested with moustache detector

After applying our moustache detector on manipulated images revealed that the detector faltered under extreme conditions (e.g. if the alterations were either too much or too little) and demonstrated an inability to identify various types of moustaches. This limitation likely stems from the methodology and an overreliance on one eigenface to fully encode moustache. Figure 2 shows faces modified with lighter-colored moustaches were not detected, suggesting that the chosen eigenface prioritised darker facial hair. Furthermore, subjects with thin or less prominent moustaches were incorrectly categorized, indicating that the eigenface may not be fully representative of all moustache variations. Conversely, these results might also reinforce the detector's robustness, as the manipulated images lacked the convincing quality required to accurately simulate the presence of moustaches.

In summary, the detector can be reliable provided that conditions remain consistent between test images and the feature being detected is also consistent from image to image. The detector is robust however prone to variations in camera angles, lighting, and facial hair characteristics, these all are liable to result in erroneous detections.

Conclusion

A technical report in PDF format of approximately **10 pages**. The report must contain an introduction, separate sections for Parts I and II (details below), **conclusions**, and **references**.